S1.5 Ideal gases

Ideal gas equation

Learning outcomes

By the end of this section you should be able to solve problems relating to the ideal gas equation.

Can samples of gas with different quantities of particles occupy the same space? Can we calculate the volume by measuring mass, temperature and pressure, using theoretical calculations in advance for practical applications? In this section, we explore the mathematical equations that relate pressure, volume, temperature and amount of moles of an ideal gas.

Practical skills

Tool 3: Mathematics — Using units, symbols and numerical values

In this section, you will be using mathematical tools in carrying out calculations with the ideal gas equation. An important aspect will be that of using the correct SI units in all calculations and being able to round all answers to the correct number of significant figures.

Combined gas law

We have seen that volume is proportional to temperature, volume is inversely proportional to pressure, and pressure is directly proportional to temperature. We were able to write the following relationships:

$$P_1V_1 = P_2V_2$$

$$\frac{P_1}{T_1} = \frac{P_2}{T_2}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

In these relationships, 1 and 2 are two different conditions of temperature, pressure and volume of a sample of gas.

These relationships can all be combined into one expression, the combined gas law, in **Table 1**.

Equation	Symbols
$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$	P_1 = pressure
	P_2 = pressure
	V_1= volume
	V_2 = volume
	T_1 = temperature
	T_2 = temperature

Study skills

When using the combined gas law $\left(\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}\right)$ you need to be careful with the units you use. Temperature should be in kelvin (K), but the units of pressure and volume can be any, as long as they are the same for both sides of the equation.

To convert from °C to K, you should add 273.15; for example, 25 °C is equal to 298.15 K.

Worked example 1

When the absolute temperature of a sample of an ideal gas at 27 °C is doubled the volume increases from 3.00 dm^3 to 6.00 dm^3. What will be the new temperature of the gas (in °C) if the pressure remains constant?

Solution steps
Step 1: Write out the values given in the question.
Step 2: Write out the equation.

Solution steps
Step 3: Rearrange the equation to make T_2 the subject.
Step 4: Substitute the values given.
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.

Worked example 2

A 2.20 dm^3 gas sample at 256 K and pressure of 100 kPa is cooled down to a temperature of 100 K while compressing it to 80.0 kPa. Calculate the new volume of the gas.

Solution steps	C
Step 1: Write out the values given in the question.	V P T P T
Step 2: Write out the equation.	
Step 3: Rearrange the equation to make V_2 the subject.	V
Step 4: Substitute the values given.	V
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	V

Ideal gas equation

The combined gas law suggests that the ratio of the product of pressure and volume to the temperature is equal to a constant:

$$\frac{PV}{T} = \text{constant}$$

We also saw that volume and moles of gas are directly proportional, and therefore the ratio of volume to moles is equal to a constant:

$$\frac{V}{n} = \text{constant}$$

The gas laws can be combined with Avogadro's law (see section S1.4.6 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/avogadros-law-of-combining-volumes-id-44265/)) to give the ideal gas equation shown in **Table 2**.

Table 2. Ideal gas equation.

Equation	Symbols	Units
$PV = nRT$	P is the pressure	pascals (Pa)
	V is the volume	cubic metres (m^3)
	n is the amount of gas	moles
	R is the universal gas constant $(8.31\ \text{J K}^{-1}\ \text{mol}^{-1})$	joules per kelvin per mo $(\text{J K}^{-1}\ \text{mol}^{-1})$
	T is the absolute temperature	kelvin (K)

In this equation, the relationships discussed in section S1.5.3 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/molar-volume-of-an-ideal-gas-and-gas-laws-id-44352/) are represented mathematically.

Study skills

The value of the gas constant R, the ideal gas equation, and the combined gas law, are given in section 1 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/some-relevant-equations-id-45103/) and section 2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/physical-constants-id-45104/) in the DP chemistry data booklet.

When using the ideal gas equation, be careful to use the correct units:

- Pressure, P, must be in Pa
- Volume, V, must be in m^3
- Amount in mol, n, this can be calculated using $n = \frac{m}{M}$, where m is the mass in grams and M is the molar mass.
- Temperature, T, must be converted to kelvin (K); if °C is given, add 273.15

It should also be noted that it is possible to use kPa for the pressure if the volume is given in dm^3.

The ideal gas equation can be rearranged to calculate the amount (in mol) of a gas (n) in **Table 3**.

Table 3. Equation for the amount of a gas

Equation	Symbols	Units
$n = \frac{PV}{RT}$	n = amount of gas	moles (mol)
	P = pressure	pascals (Pa)
	V = volume	cubic metres (m^3)
	T = temperature	kelvin (K)
	R = universal gas constant $(8.31\ \text{J K}^{-1}\ \text{mol}^{-1})$	joules per kelvin per mol $(\text{J K}^{-1}\ \text{mol}^{-1})$

Worked example 3

Calculate the moles and mass in grams of carbon dioxide that occupies 20.0 cm^3 at 25.0 °C and 100.0 kPa.

Solution steps	Cal
Step 1: Write out the values given in the question.	$V =$ $T =$ $P =$

Solution steps	Cal
Step 2 Convert the volume to m^3 and pressure to Pa.	$\frac{2}{100}$ $P =$
Step 3: Write out the equation.	$n =$
Step 4: Substitute the values given.	$n =$
Step 5: Substitute the values given.	$n =$
Step 6: Write out the equation to convert moles to mass.	$m =$
Step 7: Substitute the values.	$m =$
Step 8: State the answer with appropriate units and the number of significant figures used in rounding.	$m =$

What if you didn't know the identity of a gas but you were able to make a measurement of mass? Then you can calculate molar mass (M).

In subtopic S1.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43114/), we have learned that:

$$n \text{ (in moles)} = \frac{m \text{ (g)}}{M \text{ (g mol}^{-1}\text{)}}$$

Substituting n with $\frac{m}{M}$, we obtain:

$$\frac{m}{M} = \frac{PV}{RT}$$

Rearranging the equation for M, we obtain the equation for molar mass shown in **Table 4**.

Table 4. Equation for molar mass.

Equation	Symbols	Units
$M = \frac{mRT}{PV}$	M = molar mass	grams per mole (g mol⁻
	m = mass of gas	grams (g)
	P = pressure	pascals (Pa)
	V = volume	cubic metres (m^3)
	T = temperature	kelvin (K)
	R = universal gas constant $(8.31\ \text{J K}^{-1}\ \text{mol}^{-1})$	joules per kelvin per mo $(\text{J K}^{-1}\ \text{mol}^{-1})$

Worked example 4

Calculate the molar mass of a gas if a 500.0 cm^3 sample at 20.0 °C and 101.325 kPa has a mass of 0.666 g.

Solution steps	Calculat
Step 1: Write out the values given in the question.	$V = 50($ $T = 293$ $P = 101$ $m = 0.6$
Step 2: Convert the volume to m^3 and pressure to Pa.	$\frac{500}{1000000}$ $P = 101$
Step 3: Write the equation.	$M = \frac{m}{I}$
Step 4: Substitute the values given.	$M = \frac{0}{}$
Step 5: State the answer with appropriate units and the number of significant figures used in rounding.	$M = 32$

Practical skills

Tool 3: Mathematics — Using units, symbols and numerical values

In this section, you are applying mathematical tools to determine variables like (mass and volume) from data as accurately as possible.

International Mindedness

Units are not the same worldwide. For example, the Pascal (Pa) is the SI unit for pressure; nonetheless, several other units continue to be widely used worldwide by both scientists and non-scientists. They include atmosphere (atm), mm Hg (millimetres of mercury), Torr, bar, and psi (pounds per square inch). Due to its close proximity to 1 atm and the fact that many weather maps display pressure in millibars, the bar (1.00×10^5 Pa) is currently an often used unit. Similarly, even though the litre (dm^3) is far more frequently used, the SI unit for volume is the m^3. This is why in science we use the SI system of units. It allows universal access to science understanding.

Practical use of the ideal gas equation:

The molar mass of a gas such as butane (C_4H_{10}) can be determined experimentally using the ideal gas equation. In the experimental procedure shown in **Figure 1**, a known volume of gas from a cigarette lighter is collected in a gas syringe. The change in mass of the lighter is also recorded which gives the mass of gas collected. Together with the pressure and the temperature, the molar mass of the gas can be determined.

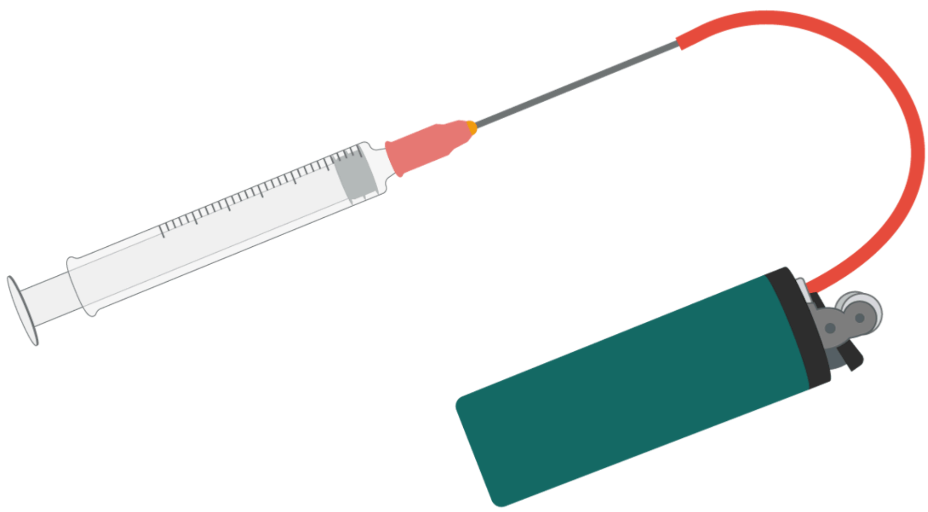

Figure 1. Collecting butane gas in a gas syringe.

More information for figure 1

Worked example 5

50.0 cm^3 of gas from a lighter was collected in a gas syringe. The change in mass of the lighter was recorded as 0.115 g. The temperature and atmospheric pressure at which the gas was collected are 298 K and 96.931 kPa respectively. Using this data, determine the molar mass of the gas.

Solution steps	Calculations
Step 1: Write out the values given in the question.	m is the change in m $(\text{in g}) = 0.115\ \text{g}$ R is the universal gas $(8.314\ \text{J K}^{-1}\ \text{mol}^{-1})$ P is the pressure of th measuring cylinder $=$ V is the volume of ga measuring cylinder 5(
Step 2: Write out the equation.	$M = \frac{mRT}{PV}$
Step 3: Substitute the values given.	$M = \frac{0.115 \times 8.31 \times}{96931 \times 0.00}$
Step 4: State the answer with appropriate units and the number of significant figures used in rounding.	$M = 58.8\ \text{g mol}^{-1}$

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:**
  - Thinking skills
    - Being curious about the natural world
    - Asking questions and framing hypotheses based upon sensible scientific rationale
    - Designing procedures and models

        - Applying key ideas and facts in new contexts
        - Engaging with, and designing linking questions
    - Social skills —Working collaboratively to achieve a common goal
- **Time required to complete activity:** 20 minutes.
- **Activity type**: Group activity

Instructions:

In a group of 2—3:

1. Find an unused box or container in your house or classroom, like a shoe box.
2. Use a ruler to record measurements that will help you calculate the volume.
3. Consult a local weather source to determine the air pressure and the daily high and low temperatures for that day.
4. Calculate the number of moles and molecules of oxygen that could fit in the box at the warmest and coldest part of the day.

Answer the following questions:

- What variables do you need to determine?
- How can you obtain them?
- What calculations do you need to perform?
- Which unit conversions do you need to apply?

Do not forget to list all resources used.